



RESEARCH REPORT



Chinese Quantum Computing Companies: Detailed Industry Analysis

20 Company Deep Profiles · Hardware, Software, QKD & Satellite

March 2026 · Prepared by Paul Oh

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\$15B

China Govt Quantum
Investment (largest
globally)

10,000km

World's Largest QKD
Network · 17 Provinces
80 Cities

#1

Global Quantum
Research Papers (since
2022)

2030

Space-Ground
Integrated Quantum
Network Goal

Executive Summary

China is the world's second-largest quantum computing ecosystem and its most strategically directed. While the United States leads in gate-model quantum hardware and commercial cloud access, China has invested approximately \$15 billion in quantum technology—the highest single-country government commitment globally—and has achieved unrivalled leadership in quantum communications: the world's largest QKD fibre network (10,000+ km, 145 backbone nodes, 17 provinces, 80 cities), the world's first quantum satellite (Micius, 2016), and the world's first real-time quantum microsatellite (Jinan-1, demonstrated March 2025 with 12,900km China-South Africa QKD). No other country has deployed quantum technology at national infrastructure scale.

China's quantum ecosystem is structurally distinct from every other nation in this report series. Where the US is dominated by private companies (IBM, Google, IonQ), France and Germany by government-backed startups (PASQAL, planqc), and Canada by academic spinouts (Xanadu, Photonic), China's quantum development is state-governed at the highest level: the 14th and 15th Five-Year Plans explicitly designate quantum technology as a national strategic priority; the Chinese Academy of Sciences (CAS) and University of Science and Technology of China (USTC) anchor the research pipeline; and state-owned enterprises (China Telecom Quantum Group, QuantumCTek) serve as the commercialisation and deployment layer. Private companies like Origin Quantum, SpinQ, and QBoson operate as 'intermediaries between public research labs and state clients'—a model fundamentally different from Western quantum ventures.

The defining technical achievements of China's quantum programme: USTC's Zuchongzhi-3 (105 qubits, 10^{15} times faster than classical supercomputers on random circuit sampling); Jiuzhang-3 (255-photon, 10^{16} quantum supremacy, photonic); Tianyan-504 (504-qubit Xiaohong chip, China's domestic qubit record, December 2024); Origin Wukong (72-qubit cloud platform, 37 million visits from 60+ countries); Jinan-1 microsatellite (March 2025, 12,900km QKD, quantum-encrypted 5G calls). In late 2025 and early 2026, a new wave of Q4 funding hit trapped-ion (Hyqubit), neutral-atom (MatriQ, CAS Cold Atom), photonic (QBoson, TuringQ), and QEC-focused (LogicBit) startups—China's first coordinated multi-modality hardware investment surge.

For Korea, China's quantum ecosystem is simultaneously a strategic threat and a critical reference point. China's 10,000+ km QKD network directly threatens Korean financial and government communications security. China's quantum cloud platforms (Tianyan, Wuyue) are already accessible to Korean researchers—and through them, Chinese quantum intelligence gathering is possible. At the same time, China's USTC-CAS-Origin Quantum model of state-anchored quantum commercialisation provides Korea with a directly applicable template for its own K-Moonshot quantum programme. The NLQIS Hefei 'Quantum Avenue' cluster model is Korea's Daedeok Innopolis quantum ambition—with a 10-year head start.

Chinese Quantum Ecosystem: Company Overview

China's quantum ecosystem concentrates in three clusters: Hefei/Anhui (USTC, CAS, Origin Quantum, QuantumCTek, CIQTEK — 'Quantum Avenue'), Beijing (Zhongguancun: Hyqubit, QBoson, ArcLight, Huawei, Alibaba DAMO), and Shanghai (photonic and neutral-atom startups). The Hefei cluster is the most productive: its proximity to USTC and CAS provides a

shared talent pool and immediate research-to-commercialisation pipeline that no other Chinese city can replicate.

Quantum Hardware — Superconducting

Company/Org	Technology	Type	Key Highlight
Origin Quantum ★	72Q Wukong SC · full-stack · USTC spinout	Private (~¥600M+)	72Q cloud 37M+ visits · IPO in process ¥6.9B
China Telecom QG ★	Tianyan cloud: 880Q cluster · Tianyan-504	SOE (China Telecom)	504Q Xiaohong Dec 2024 · Zuchongzhi-3 commercial Oct 2025
QuantumCTek ★	QKD hardware · Tianyan QC · EZ-Q fridge	Listed (STAR Market)	China Telecom controlling Jan 2025 · US Entity List
USTC / CAS ★	Zuchongzhi-3 105Q · Jiuzhang-3 photonic	Gov't Research	10 ¹⁵ SC supremacy · Pan Jianwei · PRL cover Dec 2024
LogicBit	QEC logical qubits · Zhejiang U spinout · SC	Pre-A (2025)	100Q chip · Nature paper · distance-7 surface code
Coherent Technology	Superconducting QC startup · Beijing	Private	Beijing QT Industry Conference Aug 2025 representative

Quantum Hardware — Photonic, Ion Trap, Neutral Atom

Company/Org	Technology	Type	Key Highlight
QBoson ★	Photonic QC · Beijing · first dedicated photonic fab	Private	China Mobile Wuyue cloud · bank scheduling · Beijing hub
TuringQ ★	Photonic QC · integrated silicon photonics	Private	TQI Top 80 2025 · CMOS foundry route · Guizhen sister co.
Hyqubit ★	Trapped-ion QC · Beijing	Private (2025 rd)	Series A+ 2025 · Beijing quantum hub anchor · ion trap
SpinQ ★	NMR educational QC + SC industrial · Shenzhen	Series B (2025)	40+ countries 200+ unis · Series B CCB Private Equity
CAS Cold Atom	Neutral atom QC · Wuhan Optics Valley	Gov't spinout	State-backed neutral atom · Wuhan hub anchor
MatriQ	Neutral atom QC · Shanghai · USTC-linked	Private	Shanghai neutral atom · USTC alumni founding team

Quantum Communications — QKD & Networks

Company/Org	Technology	Type	Key Highlight
China QCN ★	10,000km+ fibre QKD: 145 nodes, 80 cities, 17 provinces	Gov't (NDRC/CAS)	World's largest QKD network · Beijing-Shanghai trunk + satellite
Micius + Jinan-1 ★	Quantum satellite QKD · 2016+2022 · 12,900km March 2025	Gov't (CAS/USTC)	First quantum sat · real-time microsatellite QKD · QKD 5G
Qasky	QKD hardware manufacturer · Wuhan	Private	QuantumCTek competitor · Wuhan Optics Valley
Guoke Quantum	QKD metropolitan network operator	Private	City-scale QKD network deployment

Quantum Software, Cloud & Big Tech

Company/Org	Technology	Type	Key Highlight
Baidu Quantum ★	Paddle Quantum SDK · Qianshi QC · quantum cloud	Public (NASDAQ)	Paddle Quantum open-source ML · Qianshi superconducting QC
Alibaba DAMO ★	Quafu quantum cloud · quantum algorithm R&D	Public (NYSE)	Quafu cloud platform · aligns with national strategy
Huawei ★	HiQ quantum SDK · PQC migration · cloud QC	Private (Huawei)	HiQ Fermion simulator · PQC for telco · Kirin quantum
ArcLight Quantum	Quantum-inspired optimisation · Beijing	Private	Beijing QT hub · finance · logistics quantum algorithms

Quantum Sensing & Infrastructure

Company/Org	Technology	Type	Key Highlight
CIQTEK ★	Diamond NV quantum sensing · quantum instruments	STAR IPO (Dec 2025)	IPO ¥11.7B implied · NV-center magnetometers · spectrometers
EZ-Q Fridge	Domestic dilution refrigerator · Origin Quantum+USTC	Commercialising	450μW at 0.1K · reduces US export control dependency
NLQIS Hefei ★	National Lab for Quantum Information Science (Anhui)	Gov't (NDRC)	USTC campus · Quantum Avenue cluster anchor · CAS affiliated
MIIT / 15th FYP ★	Quantum as 'new economic growth point' 2026–2030	Gov't (MIIT/NDRC)	Commercialisation directive · industry-of-the-future designation

★ = Detailed company profile included · 60+ active quantum technology organisations across China (March 2026)

Market Overview

\$15B China Govt QT Investment — 4x USA public quantum spending	504Q Tianyan-504 Xiaohong Chip (China domestic record, Dec 2024)	12,900km Jinan-1 Microsatellite QKD: Beijing to Stellenbosch (Mar 2025)	2026–30 15th Five-Year Plan: Quantum = New Economic Growth Point
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China's Quantum Strategy: State-Governed, Nationally Deployed

China's quantum strategy differs from every other nation in this series in one fundamental way: it is not primarily a research programme or a startup ecosystem—it is a national infrastructure deployment. The Beijing-Shanghai quantum trunk line, operational since 2017. The Micius satellite, launched 2016. The 10,000km China Quantum Communication Network, operational across 80 cities. These are not technology demonstrations or pilot programmes—they are live operational networks used by Chinese government agencies, state-owned banks, and critical infrastructure operators today. No other country has deployed quantum technology at this scale or maturity.

15th Five-Year Plan (2026–2030): Quantum technology explicitly designated as a 'new economic growth point'—the highest-priority framing in Chinese industrial policy. The 14th FYP (2021–2025) focused on catching up; the 15th FYP is explicitly about quantum leadership. MIIT and NDRC directives in 2024–2025 mandate commercialisation of quantum as an 'industry of the future.'

USTC–CAS–Industry Pipeline: China's quantum research output since 2022 exceeds the United States in annual paper count (Merics). USTC's Pan Jianwei group has produced Jiuzhang (2020), Zuchongzhi series (2021–2025), and all major Chinese quantum supremacy demonstrations. CAS spinouts Origin Quantum (superconducting) and QuantumCTek (QKD) are the commercialisation vehicles for this research. The USTC–CAS–Hefei provincial government model creates an end-to-end pipeline from Nobel-level physics to national infrastructure deployment.

US Entity List Effect: QuantumCTek, several CAS institutes, and quantum supply-chain firms have been added to the US Entity List—restricting US technology acquisition. The effect, paradoxically, has been to accelerate China's domestic quantum supply chain. The EZ-Q domestic dilution refrigerator (Origin Quantum + USTC, 2024) is the most visible example: China is replacing Oxford Instruments and BlueFors with domestic alternatives at crash programme speed.

Carrier-Operated Quantum Cloud: China Telecom's Tianyan platform and China Mobile's Wuyue platform represent a model unique globally: national telcos operating quantum cloud services as infrastructure policy, not as commercial SaaS. Government agencies, state banks, and utilities are the anchor customers—ensuring revenue without enterprise sales cycles. The Tianyan platform alone has received 37 million visits from users in 60+ countries since its

November 2023 launch.

Competitive Landscape Summary

Company	Technology	Status	Funding/Scale	Key Partners
Origin Quantum	SC QC (72Q Wukong)	IPO in process	¥600M+ Series B	USTC, CAS, CTQG
China Telecom QG	Tianyan QC cloud	Operational	SOE >CNY10B plan	CAS, QuantumCTek
QuantumCTek	QKD hardware	STAR Market listed	~CNY3B market cap	China Telecom, CAS
USTC/CAS	Zuchongzhi+Jiuzhang	Research QC	Gov't national	Origin Q, CTQG
SpinQ	NMR+SC QC (edu+ind)	Series B 2025	100M+ RMB	200+ unis 40 countries
QBoson	Photonic QC	Private 2025 rnd	Private	China Mobile, banks
Hyqubit	Trapped ion	Series A+ 2025	Private	Beijing cluster
CIQTEK	Quantum sensing	STAR IPO Dec 2025	¥11.7B implied	CAS, gov't labs
Baidu Quantum	Paddle Quantum SDK	Public (NASDAQ)	Baidu corporate	BUAA, CASIA
Alibaba DAMO	Quafu cloud	Public (NYSE)	Alibaba corporate	CAS, Zhejiang U

Company Profiles

1. Origin Quantum

Company	Origin Quantum Computing Technology Co., Ltd. (本源量子) — Hefei, Anhui
Founded	2017 — USTC spinout; CEO Professor Guo Guoping (USTC quantum computing lab)
Technology	Full-stack superconducting QC: chip design → dilution fridges → control systems → cloud platform
Key System	Origin Wukong: 72-qubit superconducting QC (198-qubit chip: 72 working + 126 coupler qubits)
Cloud	Origin Quantum Cloud: 37M+ visits, 60+ countries, 339,000+ tasks executed; fine-tuned 1B-param AI model
IPO	In process at ~¥6.9 billion valuation · Series B ~¥600M+ (Guo Guoping: 'China's IBM')
Patents	#1 quantum computing patents in China · #6 globally; first domestic quantum chip production line
Korea	Samsung SDI materials, LG Chem chemistry, SK Hynix process — quantum chemistry cloud users

Technology

Origin Quantum is China's most complete quantum computing company: the only Chinese firm with a full vertical stack from superconducting chip fabrication through dilution refrigerators, control electronics, operating system (Origin OS), and cloud platform. The Wukong chip's 72-working-qubit, 126-coupler architecture mirrors IBM's tunable coupler approach—enabling dynamic inter-qubit coupling adjustment to reduce crosstalk at scale. Origin Quantum holds China's first quantum chip production line and developed China's first quantum computer operating system, making it uniquely capable of full-stack system integration without US supply chain dependencies.

The Origin Quantum Cloud's 37 million visits from 60+ countries—with US users among the most active—demonstrates that a Chinese quantum cloud platform can attract global scientific use even amid geopolitical tensions. In April 2025, Origin Wukong became the first quantum computer to fine-tune a billion-parameter AI model, demonstrating a pathway to quantum-AI hybrid computing that directly challenges NVIDIA's GPU dominance in AI training workloads.

Recent Milestones (2024 — March 2026)

- Origin Wukong 37M+ global visits (2025): 60+ countries, 339,000+ quantum computing tasks across finance, biomedicine, materials
- Billion-parameter AI model fine-tuning on Wukong (April 2025): first quantum-AI hybrid at this scale globally
- Breast cancer screening deployment: first Chinese quantum computer application in clinical medical imaging
- EZ-Q domestic dilution refrigerator (2024): Origin Quantum + USTC Anhui Engineering Centre — 450μW at 0.1K, eliminating Oxford Instruments dependency
- IPO process: ~¥6.9B valuation · China's largest quantum hardware company pre-public offering

Strategic Position for Korea

Origin Quantum's full-stack sovereign QC model—domestic chip fabrication, domestic dilution fridges, domestic cloud—is the template Korea's K-Moonshot programme is trying to build. Samsung, LG Chem, and SK Hynix are natural Origin Wukong Cloud users for quantum chemistry and materials simulation. Critically, Origin Quantum's IPO trajectory and USTC lineage make it a natural comparison benchmark for Korean conglomerates evaluating quantum investments: if Origin Quantum reaches ¥6.9B valuation with 72 qubits and 37M cloud users, what is the equivalent Korean quantum company worth?



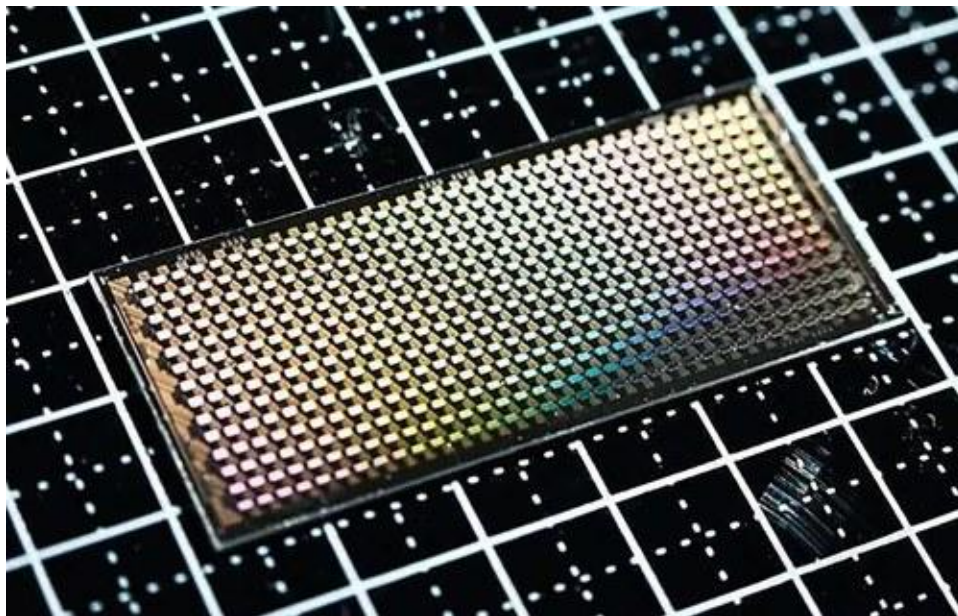
Origin Quantum Wukong chip package — 72-qubit superconducting quantum processor with 126 coupler qubits, Origin Quantum logo on aluminium housing (Hefei, 2024)

2. China Telecom Quantum Group (CTQG)

Organisation	China Telecom Quantum Group — Hefei, Anhui (wholly-owned China Telecom subsidiary)
Established	SOE quantum division; became QuantumCTek controlling shareholder January 2025
Platforms	Tianyan: 880-qubit SC cluster (24Q + 2×176Q + Tianyan-504) · China Mobile Wuyue: parallel carrier QC cloud
Key System	Tianyan-504: 504-qubit Xiaohong chip (December 2024) · co-developed with CAS + QuantumCTek
Zuchongzhi	Zuchongzhi-3 commercial deployment (October 2025): 105Q, $10^{15}\times$ faster than classical supercomputer
Investment	CNY 10 billion+ planned quantum programme investment across computing + communications
Korea	CTQG Tianyan model = K-Cloud: state telco operating quantum cloud for government/financial sector

China Telecom Quantum Group represents a structural innovation in quantum computing deployment that no Western country has replicated: a national telecommunications carrier operating quantum computing as infrastructure policy. By acquiring QuantumCTek (January 2025) and consolidating quantum computing (Tianyan platform) and quantum communications (QKD backbone) under a single SOE entity, CTQG becomes the layer through which all Chinese government, military, and financial quantum access flows. The Tianyan platform's 37 million visits demonstrate that carrier-grade quantum cloud can achieve scale without enterprise sales cycles—government agencies and state banks are the anchor customers by policy, not commercial negotiation.

- Tianyan-504 launch (December 2024): 504-qubit Xiaohong chip, China's domestic qubit record, IBM-comparable fidelity metrics
- Zuchongzhi-3 commercial deployment (October 2025): 105Q, 10^{15} faster on random circuit sampling than Frontier supercomputer
- Tianyan 880-qubit cluster (July 2025): 24Q + two 176Q machines + Tianyan-504, total 880 effective qubits
- QuantumCTek acquisition (January 2025): QKD hardware + QC cloud consolidated into single SOE entity
- Korea model: KT or SKT operating quantum cloud as national infrastructure following CTQG template



Xiaohong 504-qubit superconducting quantum chip — China's domestic qubit record, developed by CAS Center for Excellence in Quantum Information and QuantumCTek (December 2024)

3. QuantumCTek

Company	QuantumCTek Co., Ltd. (国盾量子) — Hefei, Anhui (STAR Market: 688027)
Founded	2009 — USTC / Pan Jianwei spinout; only pure-quantum listed company on Chinese exchange
Technology	QKD hardware manufacturer: BB84 transmitters/receivers, quantum random

Network	number generators, QKD network management Hardware backbone of China's 10,000+km national QKD network · 20 metropolitan QKD rings
Control	China Telecom acquired controlling stake January 2025 (QuantumCTek + CTQG QC consolidated)
US Entity	Added to US Commerce Department Entity List 2021 — banned from US technology sourcing
EZ-Q	Co-developer of EZ-Q domestic dilution refrigerator (2024) · reduces foreign dependency
Korea	Threat model: QuantumCTek QKD threatens Korean financial cryptography infrastructure

QuantumCTek is the world's largest deployed QKD hardware company—the backbone of every metropolitan and inter-city quantum-encrypted network in China. Founded as a Pan Jianwei spinout from USTC in 2009, QuantumCTek's commercial history predates every Western QKD company in volume deployment. Its listing on Shanghai's STAR Market gave it the first pure-quantum public market valuation in China. The January 2025 acquisition of control by China Telecom Quantum Group transforms QuantumCTek from an independent quantum hardware company into the hardware arm of China's national quantum infrastructure SOE.

For Korea, QuantumCTek represents the direct threat and the reference model simultaneously. The threat: QuantumCTek's QKD systems are operational in Chinese government and financial institutions whose counterparts in Korea remain protected only by RSA/ECC encryption vulnerable to quantum attack. The reference model: Korea's own national quantum network programme—led by ETRI and KT—should be benchmarked against QuantumCTek's operational 10,000km network, not theoretical plans.

- 10,000+km QKD backbone: QuantumCTek hardware powers all 145 backbone nodes, 20 metropolitan networks
- STAR Market listing: first pure-quantum public company in China · sustained revenue from government procurement
- EZ-Q dilution refrigerator (2024): QuantumCTek + Anhui Engineering Centre · first Chinese domestic QC cooling
- US Entity List impact: accelerated domestic supply chain · German SÜSS MicroTec lithographic systems remain key dependency
- China Telecom control (Jan 2025): QKD+QC consolidated under SOE for national security infrastructure

4. USTC · CAS · Pan Jianwei

Institution	University of Science and Technology of China (USTC) · Chinese Academy of Sciences (CAS)
Key Scientist	Pan Jianwei — 'Father of Chinese Quantum' · Nobel-level photonics · USTC Professor
Zuchongzhi	Series: 66Q (2021), 105Q Zuchongzhi-3 (Dec 2024) · 10^{15} faster than Frontier supercomputer

Jiuzhang	Photonic: 76Q (2020, supremacy), 113Q (2021), 255-photon (2023, 10^{16}), 105Q (2025)
Satellites	Micius (2016) + Jinan-1 (2022) · March 2025: 12,900km Beijing-Stellenbosch QKD
QEC	Distance-7 surface code QEC in progress · planning distance-9 and -11
Korea	KAIST-USTC bilateral quantum research · Korean researchers access USTC photonics expertise

USTC and Pan Jianwei's group represent China's most concentrated scientific achievement in quantum technology—and one of the most productive quantum physics groups globally. Pan Jianwei is the only scientist in the world to have demonstrated quantum supremacy in two distinct hardware modalities (photonic Jiuzhang in 2020, superconducting Zuchongzhi in 2021). Zuchongzhi-3's 105-qubit, 10^{15} classical speedup result (published Physical Review Letters as a cover article, December 2024) is directly competitive with Google's Willow chip announced the same week—the first time two countries announced competing quantum supremacy claims simultaneously.

- Zuchongzhi-3 (Dec 2024): 105Q, 99.90% single-qubit fidelity, 99.62% two-qubit fidelity · 10^{15} faster than Frontier
- Jiuzhang-3 (2023): 255-photon, 10^{16} quantum supremacy · world's fastest photonic computation
- Jinan-1 microsatellite (March 2025): 12,900km Beijing-Stellenbosch real-time QKD · quantum-encrypted 5G
- Distance-7 surface code QEC: USTC pursuing systematic QEC scale-up through 9 and 11
- KAIST-USTC bilateral: Korean quantum researchers access Pan Jianwei's photonics programme through existing agreement

5. SpinQ Technology

Company	SpinQ Technology Co., Ltd. (量旋科技) — Shenzhen, Guangdong
Founded	2018 — backed by Harvard, MIT, Tsinghua academics; unique global commercial presence
Technology	NMR-based desktop QC (educational) + industrial-grade superconducting QC
Scale	200+ universities, enterprises, research institutions across 40+ countries; 2024 revenue >50M RMB
Series B	Several hundred million RMB (mid-2025) · CCB Private Equity + government-backed investors
IPO	Preparing for Hong Kong or Shenzhen Stock Exchange listing within 12–18 months
Products	Gemini (2Q NMR desktop) · Triangulum (3Q) · industrial SC QC · Gemini Pro · 100Q target
Korea	KAIST, POSTECH, GIST quantum education labs · most accessible Chinese QC to Korean researchers

SpinQ occupies a position unique in the global quantum ecosystem: it is the only major Chinese

quantum company with genuine global commercial revenue at scale. Where Origin Quantum's Wukong Cloud attracts global users for free research access, SpinQ sells hardware and software to paying customers in 40+ countries. Its NMR-based desktop quantum computers—compact enough for classroom deployment, priced for university budgets—fill a gap that IBM, IonQ, and every Western competitor have left unaddressed: making real quantum hardware accessible to undergraduate physics and computer science students who cannot afford full-scale systems.

SpinQ's Series B (mid-2025) backed by CCB Private Equity—China Construction Bank's investment arm—signals state financial institution support for SpinQ's global expansion strategy. SpinQ's Huaxia Bank quantum neural network collaboration demonstrates the company's ability to bridge education hardware and commercial financial applications. For Korea, SpinQ's desktop QC systems are already deployed at Korean universities—making SpinQ the Chinese quantum company with the deepest existing Korean market penetration.

- 200+ universities in 40+ countries: world's most globally distributed quantum hardware company by institution count
- Series B CCB Private Equity (mid-2025): state bank-backed expansion · 100-qubit SC QC target end-2025
- Huaxia Bank quantum neural network: commercial quantum-AI application for financial decision-making
- Hong Kong / Shenzhen IPO preparation: would be China's second quantum hardware public listing after QuantumCTek
- Korean university deployment: SpinQ desktop QC systems used in Korean academic quantum education programmes



SpinQ industrial-grade superconducting quantum computer — dilution refrigerator tower for research-grade superconducting quantum computing (SpinQ Technology, Shenzhen 2025)



SpinQ Gemini desktop quantum computer in university laboratory — NMR-based 2-qubit system enabling hands-on quantum education at university budget (SpinQ Technology, 2025)

6. QBoson

Company	QBoson Co., Ltd. (玻色量子) — Beijing
Technology	Photonic quantum computing · world's first dedicated photonic QC manufacturing facility
Cloud	China Mobile Wuyue quantum cloud: photonic QC access alongside trapped-ion and SC systems
App	Postal Savings Bank of China + China Mobile: bank teller scheduling quantum algorithm (global optimal in ms)
Beijing	Beijing Quantum Industry Ecological Innovation Conference Aug 2025: representative enterprise
Korea	KDB, NH Bank, KB Financial: quantum-optimised bank scheduling directly applicable to Korea

QBoson is China's leading photonic quantum computing company and the operator of what it describes as the world's first dedicated photonic quantum computing manufacturing facility. Its deployment on China Mobile's Wuyue quantum cloud platform—alongside trapped-ion and superconducting systems—demonstrates photonic QC's readiness for cloud access alongside more established modalities. The Postal Savings Bank of China teller scheduling application is one of the clearest examples of near-term quantum value creation in China: a quantum computer solving a real combinatorial optimisation problem for a major state bank faster than any classical approach.

- World's first dedicated photonic QC manufacturing facility: vertical integration from chip to system
- China Mobile Wuyue cloud deployment: photonic QC alongside trapped-ion and SC on national carrier platform
- Postal Savings Bank quantum scheduling: global optimum found in milliseconds for branch teller allocation
- Beijing quantum hub leadership: confirmed representative at MIIT-aligned quantum industry conference Aug 2025

7. TuringQ

Company	TuringQ (图灵量子) — Shanghai
Technology	Integrated silicon photonic QC · CMOS foundry manufacturing route
Global	TQI Top 80 Quantum Companies 2025 — one of only two Chinese companies in the global list
Approach	Standard silicon photonics foundry for waveguides, beamsplitters, phase shifters at commercial scale
Korea	Samsung Foundry silicon photonics capability · potential CMOS photonic QC manufacturing partner

TuringQ is China's most globally recognised photonic quantum computing startup, one of only two Chinese companies named in the TQI Top 80 quantum companies globally in 2025. Its CMOS silicon photonics approach—manufacturing quantum optical circuits in standard semiconductor fabs rather than custom quantum cleanrooms—represents the same manufacturing philosophy as Quobly in France (FD-SOI) and PsiQuantum in the US (GlobalFoundries). If CMOS-foundry photonic circuits can reach fault-tolerant error thresholds, TuringQ's approach would be manufacturable without the capital-intensive dedicated quantum fabs that limit every other hardware modality.

- TQI Top 80 2025: one of only two Chinese quantum companies with global recognition alongside SpinQ
- CMOS silicon photonics: standard semiconductor foundry manufacturing for quantum optical circuits
- Samsung Foundry alignment: Korean fab capabilities create potential manufacturing partnership

8. Hyqubit

Company	Hyqubit (华翊量子) — Beijing
Technology	Trapped-ion quantum computing
Funding	Series A+ round 2025: Beijing quantum cluster investment surge
Beijing	Beijing Quantum Industry Ecological Innovation Conference Aug 2025: anchor trapped-ion representative
Korea	IonQ-KISTI model: Korean trapped-ion procurement should benchmark

against Hyqubit as Chinese alternative

Hyqubit is China's leading trapped-ion quantum computing startup—the Chinese counterpart to IonQ and Quantinuum in hardware modality. Its Series A+ in 2025 was part of Q4 2025's funding surge across all Chinese hardware modalities, signalling that Beijing's quantum cluster has reached the capital formation stage for trapped-ion alongside superconducting and photonic. For Korea, Hyqubit's development trajectory is directly relevant to Korea's planned KISTI 100-qubit IonQ deployment: as Chinese trapped-ion matures, Korea will face a choice between US (IonQ, Quantinuum) and Chinese (Hyqubit) trapped-ion technology—with significant geopolitical implications for whichever it chooses.

- Series A+ 2025: Beijing quantum cluster capital formation stage for trapped-ion
- Beijing hub anchor: confirmed alongside QBoson, ArcLight as representative Beijing quantum enterprises
- Korea relevance: Chinese trapped-ion alternative to IonQ for Korean national QC procurement decision

9. CIQTEK

Company	CIQTEK Co., Ltd. (国仪量子) — Hefei, Anhui
Technology	Diamond NV-centre quantum sensing · quantum precision instruments · electron spin resonance
IPO	STAR Market IPO approved December 2025 · ~¥11.7 billion implied valuation
Products	NV-centre magnetometers · single-qubit NMR · quantum diamond spectrometers · sensing systems
Customers	Research universities, national labs, industrial metrology · CAS affiliated
Korea	KRISS quantum metrology · Korean defense NV-center navigation (GPS-denied, North Korea jamming)

CIQTEK is China's most commercially mature quantum sensing company, with a product portfolio spanning NV-centre diamond magnetometers, quantum spectrometers, and precision instruments that are sold to research institutions and industrial metrology customers today. Its STAR Market IPO approval in December 2025 at an implied valuation of approximately ¥11.7 billion makes it the first Chinese quantum sensing company to achieve public market validation—and at a higher implied valuation than Origin Quantum's last funding round. For Korea, CIQTEK's NV-centre magnetometers are directly relevant to two applications: KRISS quantum metrology research and ROK defence GPS-denied navigation under North Korean jamming—the same application driving SBQuantum (Canada) and Quantum Brilliance (Germany/Australia).

- STAR Market IPO approved December 2025: first Chinese quantum sensing public listing · ¥11.7B implied
- NV-centre magnetometers: compact quantum sensors for precision navigation, medical imaging, industrial metrology
- KRISS collaboration: Korean quantum metrology institute is natural CIQTEK research partner

- ROK defence GPS-denied: NV quantum navigation for North Korea GPS jamming countermeasure

10. Baidu Quantum

Company	Baidu Research Quantum Computing Lab — Beijing (Baidu Inc., NASDAQ: BIDU)
Technology	Paddle Quantum: open-source quantum ML framework (PaddlePaddle integration)
QC	Qianshi: 10-qubit superconducting quantum computer (2022 launch, cloud accessible)
Cloud	Baidu Quantum Cloud: quantum computing as cloud service on Baidu Cloud platform
Research	Quantum simulation, quantum chemistry, quantum ML algorithms
Korea	Naver, Kakao cloud AI quantum integration · Paddle Quantum SDK Korean researcher adoption

Baidu Quantum represents China's attempt to embed quantum computing into the AI/cloud ecosystem—analogous to IBM's Qiskit integrating with its Watson AI platform. Paddle Quantum's tight integration with Baidu's PaddlePaddle deep learning framework creates a natural on-ramp for Chinese AI developers to experiment with quantum-enhanced ML—without switching toolchains. The Qianshi quantum computer (10 qubits, 2022) is smaller than IBM's or Google's systems, but its cloud accessibility through Baidu Cloud's massive existing user base gives it distribution scale that pure-quantum startups cannot match.

- Paddle Quantum: open-source quantum ML framework integrated with PaddlePaddle · thousands of Chinese AI researchers
- Qianshi QC: cloud-accessible 10-qubit SC system through Baidu Cloud platform
- Korean AI developer alignment: Naver HyperCLOVA and Kakao Karlo developers are natural Paddle Quantum users

11. Alibaba DAMO Quantum

Company	Alibaba DAMO Academy Quantum Lab — Hangzhou (Alibaba Group, NYSE: BABA)
Platform	Quafu: open quantum cloud platform for academic and research access
Research	Quantum algorithms, quantum simulation, superconducting QC · aligns with national strategy
Access	Quafu provides free cloud QC access to Chinese researchers and universities
Korea	Coupang, Kakao, Naver cloud infrastructure: Korean cloud giants face same quantum integration question as Alibaba

Alibaba's DAMO Academy quantum programme represents China's largest internet platform

company engaging with quantum computing as a strategic technology bet. The Quafu open cloud platform—providing free quantum computing access to Chinese academic researchers—mirrors IBM Quantum Network’s academic access model but within China’s domestic internet ecosystem. For Korea, Alibaba DAMO’s Quafu model is directly instructive for Korean cloud platforms: Naver Cloud and Kakao Cloud face identical strategic decisions about whether and when to integrate quantum computing services into their platforms.

- Quafu open cloud: free quantum computing access for Chinese academic researchers · domestic IBM Quantum equivalent
- National strategy alignment: DAMO quantum R&D coordinated with CAS and national Five-Year Plan priorities
- Korean cloud parallel: Naver Cloud, Kakao Cloud, and KT Cloud face same quantum cloud integration decision

12. Huawei

Company	Huawei Technologies Co., Ltd. — Shenzhen (private)
Quantum	HiQ quantum SDK · Kirin quantum processor research · PQC migration for telco infrastructure
HiQ	HiQ Fermion simulator: quantum chemistry simulation framework · open-source
PQC	Leading Chinese telco in NIST PQC standard integration for 5G/6G network security
Korea	Samsung Networks, LG U+ 5G infrastructure: PQC migration follows Huawei telco quantum model

Huawei’s quantum strategy has two distinct tracks. The research track (HiQ SDK, Kirin quantum processor R&D) builds long-term quantum hardware capability within Huawei’s semiconductor programme. The commercial track (PQC telco security) is immediately deployable: Huawei is the world’s largest 5G infrastructure provider, and its PQC-hardened 5G network equipment creates a direct channel to deploy quantum-safe communications across every country that uses Huawei network infrastructure. For Korea, Samsung Networks’ 5G equipment and LG U+’s Huawei network dependence create an indirect PQC migration dependency: if Huawei deploys PQC across its global network estate, Korean operators using Huawei gear inherit quantum-safe communications.

- HiQ Fermion simulator: open-source quantum chemistry framework · accessed by Chinese academic researchers
- PQC 5G/6G: Huawei integrating NIST PQC standards into telecom infrastructure · global network reach
- Korean telco dependency: LG U+ Huawei 5G equipment creates de-facto PQC deployment channel

13. China Quantum Communication Network (CN-QCN)

Programme	China Quantum Communication Network — operated by CAS + QuantumCTek + NDRC
Scale	10,000+km · 145 backbone nodes · 20 metropolitan networks · 17 provinces · 80 cities
Paper	npj Quantum Information August 2025: carrier-grade QKD networks over 10,000km · operational data
Satellites	Integrated with Micius (2016) + Jinan-1 (2022) for space-ground quantum network
Users	Chinese government agencies, state-owned banks, utilities: real operational quantum security
Korea	Korea National Quantum Network: CN-QCN is the operational benchmark and threat reference

The China Quantum Communication Network is the world's only operational large-scale quantum-encrypted communications infrastructure. The August 2025 Nature npj Quantum Information paper documenting CN-QCN's carrier-grade deployment across 10,000+km, 145 backbone nodes, and 80 Chinese cities represents the first peer-reviewed scientific documentation of a national quantum communication network at operational scale. This is not a research demonstration—Chinese government agencies, state banks, and utilities are using QKD for real secure communication today.



China quantum satellite-to-ground QKD concept — Micius (2016) and Jinan-1 (2022) microsatellite transmitting quantum-encrypted keys between satellite and terrestrial ground stations across intercontinental distances

- 10,000+km operational: carrier-grade QKD documented in Nature npj Quantum Information (August 2025)
- 145 backbone nodes, 80 cities, 17 provinces: world's most extensive deployed quantum communication network
- Jinan-1 March 2025: 12,900km Beijing-Stellenbosch real-time QKD · quantum-encrypted 5G calls demonstrated
- Korea threat: CN-QCN's intercontinental reach creates quantum communication capability against Korean government/financial networks

14. Micius + Jinan-1 Satellites

Micius	QUESS mission (2016) · 620kg · 500km LEO · world's first quantum satellite
Jinan-1	Quantum microsatellite (July 2022) · ~100kg · 6x higher photon frequency than Micius
Record	March 2025: 12,900km Beijing-Stellenbosch QKD · first Northern-Southern Hemisphere real-time QKD
5G	Jinan-1 + two high-throughput satellites + 5G network: first quantum-encrypted 5G satellite calls
Pipeline	3 more satellites planned 2025 · medium-high orbit satellites for global coverage by 2030
Korea	Korea's quantum satellite programme (ETRI): Jinan-1 microsatellite model directly applicable

China's satellite quantum programme is the most advanced in the world—by a large margin. Micius was the world's first quantum satellite and remains the only satellite to have demonstrated intercontinental QKD (China-Austria 7,600km, 2017). Jinan-1, weighing one-sixth of Micius and operating with ground stations that shrank from 13 tonnes to under 100 kilograms, represents the miniaturisation step from proof-of-concept to commercial deployment. The March 2025 12,900km Beijing-Stellenbosch real-time QKD is the most significant quantum communication achievement of 2025—demonstrating that a microsatellite costing a fraction of Micius can achieve transcontinental quantum-secure communication in real time.

- Micius (2016): world's first quantum satellite · China-Austria intercontinental QKD (7,600km, 2017)
- Jinan-1 (2022): microsatellite 1/6 of Micius weight · 6x higher photon frequency · 100kg ground stations
- 12,900km QKD (March 2025): Beijing-Stellenbosch real-time · one million bits per overpass
- Quantum-encrypted 5G (2025): Jinan-1 + 5G experimental network · first satellite 5G quantum security test
- Korea space QC: ETRI quantum satellite programme should benchmark against Jinan-1 microsatellite model

15–20. Brief Profiles

15. LogicBit

Company	LogicBit (逻辑比特) — Hangzhou (Zhejiang University spinout)
Focus	Quantum error correction · logical qubits · superconducting surface code hardware
Funding	Pre-A round 2025
Korea	KAIST QEC research group: LogicBit surface code approach benchmarks against KAIST theoretical work

LogicBit is China's most technically ambitious quantum hardware startup—not competing on

qubit count but on error correction. Its Nature paper (100-qubit chip), distance-7 surface code QEC roadmap, and explicit benchmarking against Google's QEC architecture position LogicBit as the Chinese company most likely to matter in the fault-tolerant era. If the logical qubit era arrives on Google and IBM's projected timelines, LogicBit will prove more strategically consequential than Origin Quantum's larger-qubit-count but lower-fidelity approach.

16. CAS Cold Atom / MatriQ

Companies	CAS Cold Atom (Wuhan) + MatriQ (Shanghai USTC-linked) — neutral atom QC
Significance	China's neutral atom entrants · Q4 2025 funding wave · Beijing: Liangyi Wanxiang
Korea	KRISS cold atom research: Korean-Chinese neutral atom research exchange programme

China's neutral atom ecosystem is the fastest-growing quantum hardware modality in 2025–2026, with CAS Cold Atom (Wuhan), MatriQ (Shanghai), Buchou Quantum, and Beijing's Liangyi Wanxiang all emerging within a compressed timeframe. The rapid formation mirrors France's PASQAL genesis and Germany's planqc—but with China's unique advantage of USTC-trained physicists providing the founding talent pool. For Korea, neutral atom's room-temperature operation and scalability make it the modality most relevant to Korea's K-Moonshot programme—and Chinese neutral atom startups are the closest geographic partners for joint development.

17. ArcLight Quantum

Company	ArcLight Quantum (弧光量子) — Beijing
Focus	Quantum-inspired and quantum software · optimisation algorithms
Market	Finance · logistics · manufacturing optimisation using quantum and quantum-inspired methods
Korea	Samsung SDS, LG CNS enterprise software division: quantum-inspired optimisation direct parallel

ArcLight Quantum represents Beijing's quantum software layer—the Chinese equivalent of Kipu Quantum (Germany) or 1QBit (Canada) in applying quantum and quantum-inspired algorithms to real commercial optimisation problems. Its participation in the Beijing Quantum Industry Ecological Innovation Conference 2025 alongside hardware companies (Hyqubit, QBoson) confirms ArcLight as a software ecosystem anchor for Beijing's multi-modality quantum cluster.

18. NLQIS Hefei — Quantum Avenue

Institution	National Laboratory for Quantum Information Science (NLQIS) — Hefei,
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Location	Anhui 'Quantum Avenue' (量子大道) · USTC campus adjacent · dozens of quantum firms co-located
Function	National quantum research anchor · shared talent pool · research-to-commercialisation pipeline
Korea	Daedeok Innopolis + KAIST: Korea's quantum cluster development should mirror Quantum Avenue model

NLQIS Hefei and its surrounding Quantum Avenue cluster represent the most productive quantum research-to-industry pipeline in Asia. The proximity of USTC (research), NLQIS (national lab), Origin Quantum (hardware commercialisation), QuantumCTek (QKD deployment), CIQTEK (sensing), and dozens of startups within walking distance of each other creates a talent density and collaborative spillover that no government programme can manufacture artificially. For Korea, the Daedeok Innopolis quantum cluster—centred on KAIST, ETRI, and the planned National Quantum Computing Centre—is Korea's attempt at the Quantum Avenue model. Hefei has a 10-year head start; Korea's cluster must learn from Hefei's structure while moving at Korean speed.

19. MIIT / 15th Five-Year Plan

Policy	Ministry of Industry and Information Technology (MIIT) + National Development and Reform Commission (NDRC)
15th FYP	2026–2030: quantum technology as 'new economic growth point' · highest-priority framing
MIIT 2024	Directive: commercialisation of quantum as 'industry of the future' · 11.56B RMB industry target by 2025
Korea	MSIT K-Quantum strategy: Korea's 2025 quantum industrial policy should benchmark against 15th FYP scope

China's 15th Five-Year Plan quantum designation is the most significant quantum policy signal in Asia in 2026. Previous Five-Year Plans funded quantum as research excellence. The 15th FYP frames quantum as an economic growth driver—shifting the mandate from scientific achievement to industrial deployment. MIIT's 2024 commercialisation directive targets an 11.56 billion RMB (US\$1.61 billion) quantum computing industry by 2025 at 30%+ annual growth. For Korea's MSIT, the 15th FYP trajectory—how China transitions from state-funded research to commercially self-sustaining quantum industry—is the single most important quantum policy reference in Asia.

20. EZ-Q Domestic Dilution Refrigerator

Programme	EZ-Q Fridge — Origin Quantum + USTC Anhui QC Engineering Research Centre (2022 setup, 2024 production)
Specs	450 microwatts at 0.1 Kelvin · factory production (not just lab prototype) as of

	February 2024
Upgrade	1,000 microwatt version in development · targets parity with Oxford Instruments BlueFors
Meaning	US Entity List forced domestic dilution fridge: one of the last critical foreign dependencies being eliminated
Korea	Korean NQCC sovereign cryogenic supply: German kiutra + Chinese EZ-Q as alternatives to Oxford Instruments monopoly

The EZ-Q domestic dilution refrigerator is China’s most strategically significant quantum supply chain achievement. Dilution refrigerators—which cool superconducting quantum processors to millikelvin temperatures—have been a critical dependency on UK-manufactured Oxford Instruments and Finnish BlueFors equipment. The US Entity List restrictions forced a crash programme: the Anhui Quantum Computing Engineering Research Centre (established by Origin Quantum and USTC in 2022) began factory production of the EZ-Q Fridge in February 2024, delivering 450 microwatts at 0.1 Kelvin. For Korea, the EZ-Q development—alongside Germany’s kiutra ADR cryostats—provides Korea’s planned National Quantum Computing Centre with supply chain alternatives to Oxford Instruments’ global monopoly.

Implications for Korea Market Entry

China-Korea Quantum Strategic Alignment and Threat

China and Korea's quantum relationship is uniquely complex: China is simultaneously Korea's largest trading partner, its most proximate military threat, and the quantum ecosystem most directly relevant to Korea's industrial structure. BASF/BMW/Siemens (Germany) and Samsung/LG/Hyundai (Korea) share industrial DNA. But POSCO/Samsung/Hyundai and Baosteel/CATL/BYD share supply chain competition DNA—meaning China's quantum advantage in materials simulation, battery chemistry, and manufacturing optimisation is a direct commercial threat to Korea's industrial competitiveness, not just a geopolitical concern.

Korea Need	Chinese Company	Strategic Opportunity / Threat
K-Moonshot QC Benchmark	Origin Wukong 72Q (THREAT)	China's 37M-visit quantum cloud = free Korean researcher access = Chinese quantum intelligence gathering opportunity
National QKD Network	CN-QCN 10,000km (THREAT)	China's QKD network already reaches Korean border · Korea's RSA/ECC finance vulnerable
Quantum Satellite Programme	Micius + Jinan-1 (THREAT/MODEL)	ETRI quantum satellite should benchmark Jinan-1 microsatellite model · 100kg, realtime, low-cost
Battery/Materials QC	Origin Quantum + CTQG (THREAT)	CATL battery chemistry quantum advantage would directly threaten Samsung SDI/LG Energy Solution market position
Quantum Cloud Infrastructure	Tianyan + Wuyue (MODEL)	KT/SKT quantum cloud: carrier-operated model from CTQG · government anchor customers first
Domestic Supply Chain	EZ-Q Fridge (MODEL)	Korea NQCC dilution fridge: EZ-Q + kiutra (Germany) as alternatives to Oxford Instruments monopoly
QC Education Platform	SpinQ Technology (OPPORTUNITY)	Korean university labs already using SpinQ · most accessible Chinese QC · low geopolitical risk
Quantum Sensing / Defence	CIQTEK NV-center (DUAL USE)	KRISS quantum metrology partnership · but NV-center tech is dual-use for GPS-denied navigation
PQC / Network Security	QuantumCTek + Huawei (THREAT)	Korean 5G/6G networks: Huawei PQC migration + QuantumCTek QKD threat to Korean financial institutions
Quantum Avenue Model	NLQIS Hefei cluster (MODEL)	Daedeok Innopolis quantum cluster: KAIST+ETRI+KIST co-location should mirror Quantum Avenue structure

United States vs China: Comparative Positioning for Korea

Dimension	United States	China
Hardware	IBM, Google, IonQ, Quantinuum — world-leading gate-model across all modalities	Origin Wukong 72Q, Tianyan-504 504Q, Zuchongzhi-3 105Q, Jiuzhang-3 photonic — competitive supremacy results
QKD/Quantum Comms	US deprioritises QKD (NSA guidance) · no national QKD network	World leader: 10,000+km operational QKD network · Micius + Jinan-1 satellite · quantum-encrypted 5G
Cloud Access	AWS Braket Korea, Azure Quantum, IBM Quantum — commercial cloud available in Korea	Tianyan (37M visits, 60+ countries), Origin Cloud, Wuyue — open access but with data sovereignty risk
Geopolitical Risk	ITAR/EAR export controls may restrict hardware to Korea in defense contexts	Chinese-origin data, quantum intelligence gathering risk for Korean industrial/government use
Materials/Battery	Schrödinger, Good Chemistry — leading chemistry SW	Origin Quantum battery chemistry + CATL parallel quantum programme — direct Korean industry threat
Policy Model	NIST PQC standards · DARPA QBI hardware validation · private-led ecosystem	15th FYP state-led · carrier-operated cloud · state bank anchor — Korean SOE quantum template
Education QC	IBM Quantum Educator, QiskitLearning — dominant global quantum edu platform	SpinQ desktop QC in Korean universities already · most accessible Chinese QC hardware
Industry Match	Finance, pharma, energy (JPMorgan, Goldman Sachs focus)	Battery (CATL), steel (Baosteel), EV (BYD) — direct competitors to Samsung SDI, POSCO, Hyundai
Research Output	US led quantum papers 2019–2021 · most citations per paper	China #1 annual quantum paper count since 2022 · lower average citation but rising rapidly

Strategic Recommendation: Korea's China quantum strategy requires a fundamentally different framework than its US/EU quantum partnerships. With the US and European nations (Germany, France, Canada), the question is 'how do we partner?' With China, the question is 'how do we compete, protect, and selectively engage?' Korea should: (1) **PROTECT** — prioritise deployment of Korea's national QKD network to match CN-QCN's operational capability, with particular urgency for financial sector PQC migration; (2) **BENCHMARK** — use Origin Quantum's 72Q cloud, USTC's research outputs, and CN-QCN's 10,000km network as the competitive targets for Korea's K-Moonshot programme; (3) **MODEL** — adopt China's Quantum Avenue cluster model (NLQIS+USTC+Origin Quantum) for Daedeok Innopolis, the carrier-operated quantum cloud model (CTQG Tianyan) for KT/SKT, and the EZ-Q domestic supply chain approach for Korean quantum cryogenics; and (4) **ENGAGE SELECTIVELY** — SpinQ for accessible university quantum education hardware, CIQTEK for quantum sensing research, and CAS Cold Atom / MatriQ for neutral atom academic exchange, avoiding Chinese quantum cloud platforms for any sensitive Korean government or industrial workloads.

Conclusion

China's quantum ecosystem is the world's most consequential for Korea—not because it is the

most technically advanced (the US leads in gate-model hardware), but because it is the most operationally deployed, the most geographically proximate, and the most directly competitive with Korea's industrial structure. China's 10,000km QKD network is operational, protecting Chinese government and financial institutions with quantum encryption while Korean equivalents remain protected only by classical cryptography. China's USTC-CAS research pipeline has produced the world's only country with quantum supremacy demonstrations in two modalities. China's 15th Five-Year Plan designates quantum as an economic growth driver precisely because CATL, Baosteel, and BYD's quantum advantages in battery chemistry and materials simulation will directly challenge Samsung SDI, POSCO, and Hyundai's market positions.

The K-Quantum strategy imperative is clear: Korea must treat China's quantum programme as both a strategic threat requiring protective countermeasures (national QKD network, PQC migration for financial sector, quantum-safe government communications) and a policy model worth selective adoption (Quantum Avenue cluster structure, carrier-operated quantum cloud, domestic cryogenic supply chain). The NLQIS Hefei model, CTQG Tianyan platform model, and EZ-Q domestic fridge programme are the three most directly transferable Chinese quantum policy innovations for Korea's K-Moonshot implementation. China has a 10-year head start in quantum deployment. Korea's response must be faster, more focused, and operationally oriented—not just research-oriented.

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